

precog-data-intake: Automated discovery, validation, and optimized download management of Earth system model data from Earth System Grid Federation nodes.

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Summary

precog-data-intake is a Python command-line-interface (CLI) software for automated discovery, checking, validation, and download of Earth System Model outputs from Earth System Grid Federation (ESGF) archives. The software is designed for research workflows that require reproducible access to large, distributed climate-model datasets and is particularly aimed at bulk screening of Earth system archives before downstream analysis. The software builds on inherited ESGF access functionality from intake-esgf (Collier et al., 2026), while introducing interactive CLI workflows for archive interrogation, file-availability checks, grid-consistency cross-checks, temporal validation, export of search results in tabular formats, and optimised download management of shortlisted Earth System Model data products.

Statement of need

Modern Earth System science workflows frequently rely on climate-model data distributed across Earth System Grid Federation (ESGF) nodes (Cimadevilla et al., 2025). In the context of marine biogeochemistry for instance, Couple Model Intercomparison Project analyses centre on evaluating how Earth System Models simulate changes in ocean carbon, oxygen and nutrient fields and fluxes, quantifying multi-model biases and uncertainties to constrain future projections of ocean carbon inventories for example (Wilson et al., 2022). Although ESGF provides a federated infrastructure for searching and accessing these archives, practical research workflows often require additional automation to determine whether a given combination of variables, experiments, ensemble members, and grid configurations is both scientifically suitable and operationally downloadable before substantial time is spent retrieving files.

This issue is especially important for CMIP-style analyses in which researchers may need to confirm that the same Earth System Model provides paired pre-industrial, historical and ssp-scenario simulations, that multiple variables of interest are available and whether these are separately archived on compatible grids for downstream analyses; and that temporal coverage is continuous across archived files. Manual inspection of catalogue search results can become slow, repetitive, and error-prone when screening many candidate models or variables across multiple ESGF nodes.

Among its features, precog-data-intake provides an interactive workflow for shortlisting Earth System Model datasets that satisfy compound criteria across experiments and variables. The software validates temporal coverage, grid consistency, and file availability across published ESGF archives before download. Users can export search results as tabular summaries, inspect and refine shortlisted datasets, and initiate batch downloads or trigger file integrity checks on locally assigned output paths. Retrieved data are then organised into a directory structure

41 suitable for reproducible downstream analysis. This workflow is particularly useful when
42 researchers need to screen many candidate models and variables before selecting datasets that
43 are both scientifically appropriate and operationally accessible within their computational and
44 storage constraints.

45 Furthermore, by wrapping ESGF discovery and CMIP variable conventions (including less
46 familiar fields like `volcello`, `areacello`, and input validation when searching for any variable
47 of interest) behind a simple, high-level Python CLI interface, `precog-data-intake` lowers
48 the barrier for researchers new to CMIP and the ESGF ecosystem to find, load, and start
49 analysing Earth System Model outputs without the need for mastering CMIP metadata and
50 directory conventions. The simple design also enables `precog-data-intake` to be used directly
51 in terminal-based HPC environments, allowing users to run data discovery and ingestion
52 from visualization nodes and Jupyter sessions relying on interactive web interfaces or manual
53 downloads.

54 State of the field

55 CMIP data are currently distributed through the ESGF metagrid web application, which
56 provides web-based discovery and download services for CMIP5, CMIP6, CMIP6Plus, and
57 the upcoming CMIP7 phases. Recent development efforts have focused on analysis-ready,
58 cloud-optimised approaches based on in-memory object storage (Mizielinski et al., 2026),
59 with catalogue sweep tools enabling more scalable access through Python and xarray, while
60 community evaluation frameworks like ESMValTool(Righi et al., 2020) supporting integrated
61 diagnostics for benchmarking model outputs.

62 `precog-data-intake` builds on the ESGF catalogue node sweeping implementation from
63 `intake-esgf` (Collier et al., 2026), but targets a different level of the workflow: rather than
64 focusing only on data access primitives and caching data-responses to in-memory use, it
65 provides an interactive command-line environment for ample discovery, screening, validating
66 and managing downloads of published ESGF archive products that satisfy scientific constraints
67 relevant to Earth system and ocean biogeochemistry applications. The software is intended for
68 situations in which researchers need to move efficiently from broad discovery and catalogue
69 searches to a smaller, analysis-ready subset of model output that satisfies practical and scientific
70 constraints.

71 Key functionality

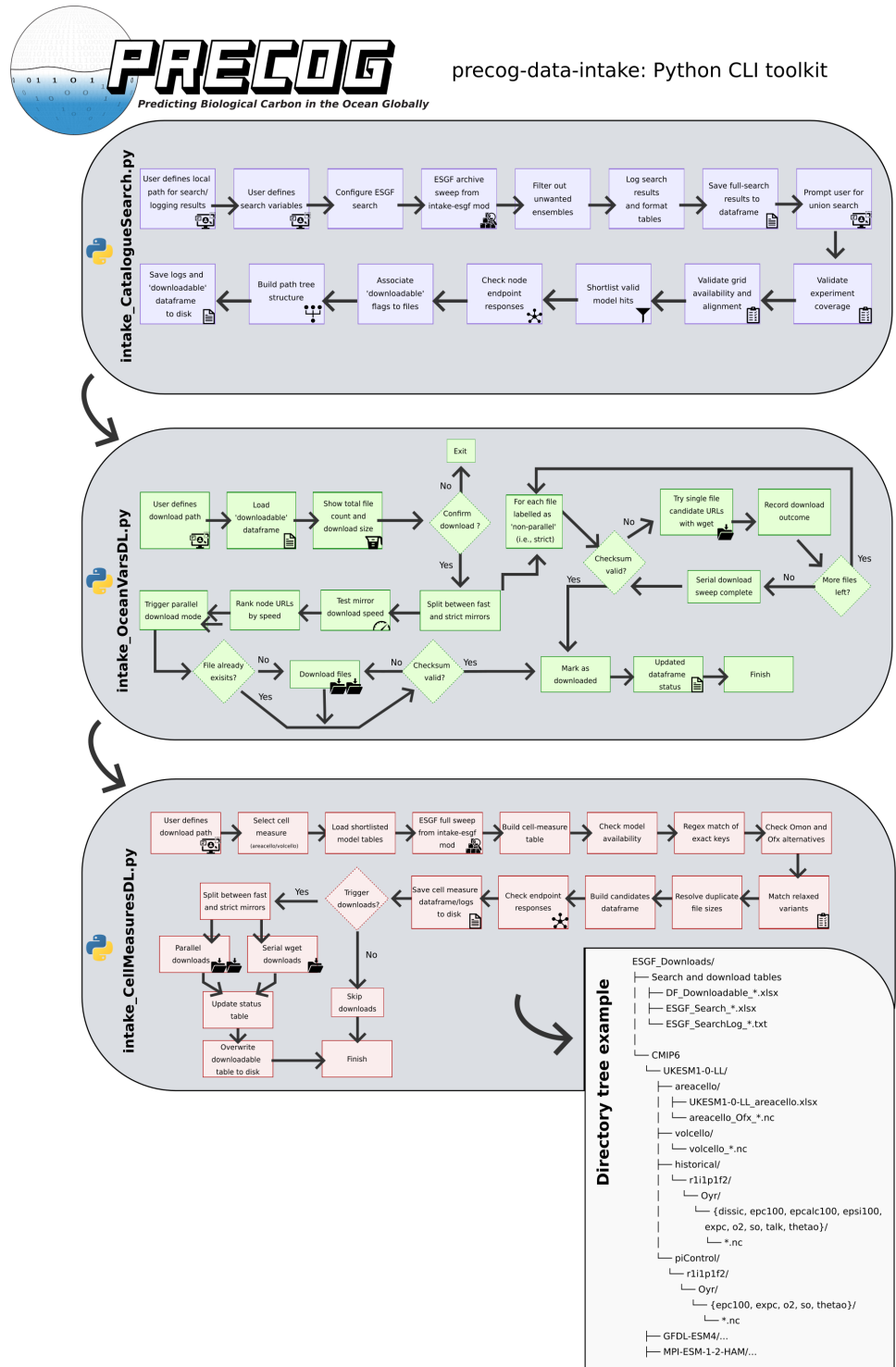
72 The software provides several features tailored to archive-scale Earth system data workflows:

- 73 ■ automated ESGF node-based searches using project, activity, experiment, frequency,
74 variable, and grid filters (inherited from `intake-esgf`);
- 75 ■ interactive user prompts for download paths and variable selections within CLI workflows;
- 76 ■ bulk screening of CMIP6 archive holdings for paired `piControl` and historical
77 availability;
- 78 ■ validation of continuity in date stamps for CMIP6 pre-industrial and historical simulations;
- 79 ■ checks for consistent availability of model output across compatible grids such as `gn` and
80 `gr`;
- 81 ■ conditional multi-variable screening for models that satisfy compound search criteria,
82 including cases where several required variables must be available simultaneously;
- 83 ■ export of tabular ESGF catalogue search results for easy inspection, search snapshot
84 logging, and reproducible shortlist generation;
- 85 ■ parallel URL pre-checks with verification of server responses and flagging of shortlisted
86 outputs as downloadable,
- 87 ■ download manager with parallel transfers that select responsive ESGF node locations
88 and include retry or integrity checks for corrupted files

- 89 ▪ in cases where the same file is available across many nodes, the fastest download option
- 90 is prioritised based on the user's connection
- 91 ▪ retrieval of ocean grid-cell measure variables such as areacello and volcello for selected
- 92 models, together with saved ESGF archive snapshots for later inspection.

93 **Software design and workflow overview**

94 precog-data-intake implements a staged workflow for archive-scale ESGF data discovery and
95 retrieval. Rather than moving directly from catalogue search to cached download, the software
96 separates archive interrogation, shortlist generation, downloadability checks, and file retrieval
97 into distinct command-line steps, allowing users to inspect and validate intermediate results
98 before proceeding. In a typical workflow (Figure 1), the user first specifies a parent download
99 directory and one or more target variables. The catalogue search stage then queries ESGF
100 holdings for matching CMIP products (CMIP6 as the default option), filters results to retain
101 scientifically relevant combinations such as paired piControl and historical simulations, and
102 exports tabular search summaries for inspection. Subsequent stages verify whether shortlisted
103 files are reachable on remote nodes, assign local destination paths, and download both target
104 variables and required supporting grid-cell measures such as areacello and volcello. This
105 staged design is intended to improve transparency and reproducibility in archive-based Earth
106 system workflows. By treating search results, validation outputs, and downloadable file lists as
107 explicit intermediate artifacts, the software supports both interactive use and later auditing of
108 dataset selection decisions.



109

110 {width=90%} Figure 1. precoc-data-intake toolkit overview and directory structure of an
111 example ESGF download. The top-level directory contains search outputs and model-specific
112 CMIP6 data organized by model, experiment, variable, and annual files.

Research impact and applications

A representative use case is the identification of CMIP6 models that simultaneously provide both pre-industrial and historical outputs for ocean biogeochemical variables such as `expc` and `epc100`, together with auxiliary or supporting variables and the associated grid-cell measures required for downstream analyses. The repository accompanying this software includes a [workflow example](#) demonstrating this type of archive screening and retrieval process.

This is particularly relevant for ocean biogeochemistry and carbon-cycle studies for example, where analyses often depend on coherent combinations of physical and biogeochemical fields rather than isolated variables (e.g., retrieval of carbonate system fields as well as ocean state physical variables, (Wilson et al., 2022)). In such cases, the time spent screening archive holdings, checking consistency, and organising downloads can be substantial, and purpose-built automation improves both efficiency and reproducibility.

Therefore, `precog-data-intake` fills a workflow gap between catalogue access and scientific analysis. The software capabilities support workflows in which data access is itself a significant part of the scientific process, particularly when analyses depend on assembling coherent, scientifically consistent, and analysis-ready ensemble subsets of Earth system model output from distributed archives.

AI usage disclosure

AI assistance was used solely to improve readability of selected software documentation (Claude Sonnet 5 by Anthropic). No AI tools were used to design, implement, or validate the software. All AI-assisted documentation was reviewed and edited by the authors, who retain full responsibility for its accuracy and content. Authors made all the core design and architectural decisions.

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